

Programmable Three-Mode Interferometer

Fock-State Inputs and PNR Output Sampling

Anchor: Three-mode passive interferometer with tunable Mach–Zehnder interferometer phases.

Purpose: Demonstrate programmable photonic interference, photon-number conservation, and phase-dependent suppression or enhancement of allowed output patterns.

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Abstract

This brief describes a compact simulator for a programmable three-mode passive interferometer. The simulator uses selectable Fock-state inputs $|n_0, n_1, n_2\rangle$, with $n_i \in \{0, 1\}$, three tunable Mach–Zehnder interferometers (MZIs), and photon-number-resolved (PNR) output measurement. The goal is to provide a simple public artifact showing how phase-programmed linear optics reshapes many-photon output probability distributions. Because the device is passive, total photon number is conserved. However, individual output patterns that satisfy photon-number conservation can still be suppressed or enhanced through interference. The simulator therefore provides a minimal demonstration of programmable photonic sampling and a component-level building block for later photonic optimization, unitary-decomposition, and photonic-integrated-circuit (PIC) translation workflows.

Core demonstration. The simulator shows that tuning three MZI phases can make certain photon-number-allowed output states disappear. These zeros are not imposed by the photon-number constraint; they arise from destructive multi-photon interference.

Background

Passive linear optics transforms optical modes while preserving total photon number. For a three-mode input Fock state,

$$|n_0, n_1, n_2\rangle_{\text{in}}, \quad n_i \in \{0, 1\},$$

the total input photon number is

$$N_{\text{in}} = n_0 + n_1 + n_2.$$

A passive interferometer implements a unitary transformation U on the optical modes. Therefore, the output distribution is supported only on patterns

$$|m_0, m_1, m_2\rangle_{\text{out}}$$

that satisfy

$$m_0 + m_1 + m_2 = N_{\text{in}}.$$

For example, an input $|1, 1, 0\rangle$ produces only two-photon output patterns:

$$|2, 0, 0\rangle, \quad |1, 1, 0\rangle, \quad |1, 0, 1\rangle, \quad |0, 2, 0\rangle, \quad |0, 1, 1\rangle, \quad |0, 0, 2\rangle.$$

This constraint makes the simulator physically transparent and keeps the output histogram compact.

The simulator uses Fock-state inputs rather than squeezed-state inputs. For passive linear optics with Fock-state inputs, output amplitudes are governed by matrix permanents of submatrices of the interferometer unitary. This is distinct from Gaussian boson sampling with squeezed inputs, where Hafnian- or Torontonian-based probability formulas arise. The present simulator therefore demonstrates the programmable interferometer layer without the additional overhead of squeezed-state simulation.

Objective

The objective is to create a small, transparent, and reusable simulator that demonstrates programmable interferometer control. The user selects input occupations n_0, n_1, n_2 , tunes three MZI phases, and observes the full PNR output probability distribution over all photon-number-conserving output patterns.

Public artifact strategy. This simulator is intentionally generic. It demonstrates component-level photonic programming without exposing proposal-specific feature-encoding, optical-scoring, or application-domain workflows.

The simulator supports three related goals:

- demonstrate how tunable MZI phases program a passive interferometer;
- show photon-number conservation in PNR output distributions;
- illustrate interference-driven suppression and enhancement of allowed output patterns.

Approach

The interferometer consists of three MZI layers acting on three optical modes:

$$(0, 1) \rightarrow (1, 2) \rightarrow (0, 1).$$

Each MZI has one tunable internal phase. The full transformation is

$$U_{\text{mesh}} = U_{01}^{(2)}(\phi_{01}^{(2)})U_{12}^{(1)}(\phi_{12}^{(1)})U_{01}^{(0)}(\phi_{01}^{(0)}).$$

The MZI convention is based on a balanced beamsplitter, an upper-arm phase, and a second balanced beamsplitter:

$$MZI(\phi) = B P(\phi) B,$$

with

$$B = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix}, \quad P(\phi) = \begin{pmatrix} e^{i\phi} & 0 \\ 0 & 1 \end{pmatrix}.$$

Up to a physically irrelevant global phase, this gives the effective real MZI block

$$M(\phi) = \begin{pmatrix} \sin(\phi/2) & \cos(\phi/2) \\ \cos(\phi/2) & -\sin(\phi/2) \end{pmatrix}.$$

This form is useful for an interactive simulator because it makes the phase dependence easy to interpret: changing ϕ continuously tunes how strongly the two modes mix.

For a selected input state $|\mathbf{n}\rangle$ and output state $|\mathbf{m}\rangle$, with

$$\mathbf{n} = (n_0, n_1, n_2), \quad \mathbf{m} = (m_0, m_1, m_2),$$

the transition probability is computed from the passive-linear-optics Fock transition amplitude

$$A_{\mathbf{m},\mathbf{n}} = \frac{\text{perm}(U_{\mathbf{m},\mathbf{n}})}{\sqrt{\prod_i n_i! \prod_j m_j!}}.$$

Here $U_{\mathbf{m},\mathbf{n}}$ is formed by repeating columns of U_{mesh} according to the input occupations and repeating rows according to the output occupations. The output probability is

$$P(\mathbf{m}|\mathbf{n}) = |A_{\mathbf{m},\mathbf{n}}|^2.$$

Only output occupations satisfying

$$\sum_j m_j = \sum_i n_i$$

are enumerated and displayed.

Simulator workflow.

1. Select input occupations $n_0, n_1, n_2 \in \{0, 1\}$.
2. Tune the three MZI phases $\phi_{01}^{(0)}$, $\phi_{12}^{(1)}$, and $\phi_{01}^{(2)}$.
3. Construct the three-mode unitary U_{mesh} .
4. Enumerate all PNR output patterns with conserved photon number.
5. Compute permanent-based transition probabilities.
6. Display the output distribution as a probability histogram.

Takeaway

The simulator demonstrates the essential behavior of a small programmable photonic interferometer. Total photon number is conserved, but the distribution over allowed output patterns is highly phase dependent. At special phase settings, some output states that are allowed by photon-number conservation can have zero probability because the corresponding multi-photon amplitudes cancel destructively.

This is the central technical message: a programmable interferometer does more than route photons independently. It reshapes a many-photon probability distribution through coherent interference. That makes the simulator useful as a lightweight public demonstration of programmable photonic sampling and as a component-level artifact supporting broader work on photonic optimization, unitary decomposition, and future PIC implementation.

Final interpretation. The three-mode simulator is a minimal bridge between optical component programming and photonic sampling. It shows how MZI phase settings become experimentally meaningful output probabilities, while keeping the model simple, auditable, and easy to extend.

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